

Jose Jathir Rubio Delgado

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Professional Summary

Mechatronics engineer with hands-on experience in industrial automation, robot programming, and CAD-driven digital twins. Strong background in scripting 3D model pipelines, PLC programming, and building autonomous robotics routines with ROS and LiDAR. Comfortable delivering repeatable, scriptable solutions, optimizing operator–robot workflows, and collaborating across R&D and manufacturing environments.

Experience

ZEISS — Oberkochen, Germany
R&D Department — Thesis

Sep 2024 – Feb 2025

- Automated CAD model processing for recurring tasks: component rotation, removal of unnecessary elements, and vertex-count optimization.
- Drove script-based control of 3D animation and rendering software to ensure robustness and reproducibility in model pipelines.
- Identified relevant CAD parameters and analyzed their impact on pose estimation and tracking performance.

ZEISS — Oberkochen, Germany
R&D Department — Internship

Apr 2024 – Sep 2024

- Implemented scripted workflows to modify and evaluate CAD models for robust tracking.
- Assessed effects of model changes to prepare robustness and accuracy tests for CAD model tracking.
- Tuned object model parameters to improve overall system performance.

Hochschule Zittau/Görlitz — Zittau, Germany
Robotics Programming (ROS)

Sep 2023 – Jun 2023

- Programmed a LIMO robot in Python/ROS to execute a complex routine: room mapping via LiDAR, autonomous navigation, and color recognition with onboard vision.

Skills

Programming: Python, C++, LabVIEW, MATLAB.

Robotics/Automation: ROS, LiDAR mapping, Robot programming (xArm), PLCs (Siemens TIA Portal).

CAD/Simulation: SolidWorks, Process Simulate, script-based CAD/animation control, CNC G-code.

Methods: Digital twin modeling, scripting for reproducible pipelines, prototyping, testing, documentation.

Languages: Spanish (Native), English (B2 – Intermediate), German (A2 – Basic).

Education

Hochschule Zittau/Görlitz — Zittau, Germany

Oct 2023 – Feb 2025

B.Eng. Automatisierung und Mechatronik (DHIK double-degree path) – one-year exchange culminating in a thesis. Note: 1.22/1 — Equivalent to GPA 3.9/4.0

Tecnológico de Monterrey — Monterrey, Mexico

Aug 2020 – Jul 2025

B.Sc. in Mechatronics Engineering – Note: 96/100 — Equivalent to GPA 3.84/4.0

Awards & Memberships

Member, DHIK Dual Degree (2023–2024). 50% Scholarship for academic talent (2020–2025).

Certifications

E-Series Application, Universal Robots (2023). E-Series Core Track, Universal Robots (2023). E-Series Pro Track, Universal Robots (2023). CSWA SolidWorks, XpertCAD (2023). Learn C++ Programming (Basic–Advanced), Udemy (2023). LabVIEW and Arduino Application Programming, Udemy (2022).